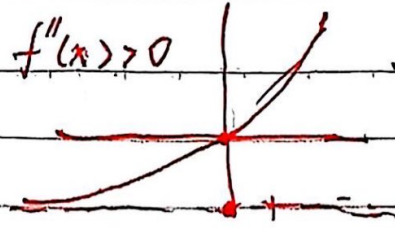


(a) $f(x) = e^x - 1$ on $\mathbb{R}$ $f''(x) > 0$
 strictly convex $\Rightarrow$ quasiconvex



(b) $f(x_1, x_2) = x_1 x_2$ on $\mathbb{R}_{++}^2$

$$\nabla^2 H(f) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \Rightarrow \begin{cases} \lambda_1 = 1 \\ \lambda_2 = -1 \end{cases}$$

eigenvalues are neither positive ~~semidefinite~~ semidefinite
 nor negative semidefinite

① f is neither convex nor concave

② Consider its sub level:

$\{(x_1, x_2) \in \mathbb{R}_{++}^2 \mid x_1 x_2 \leq \alpha\}$ is convex

Accord with $f[\lambda x + (1-\lambda)y] \geq \min\{f(x), f(y)\}$

$\therefore f$ is quasiconcave

(c) $f(x_1, x_2) = 1/(x_1 x_2)$ on $\mathbb{R}_{++}^2$

$$\nabla^2 H(f) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix} = \frac{1}{x_1 x_2} \begin{pmatrix} \frac{2}{x_1^2} & \frac{1}{x_1 x_2} \\ \frac{1}{x_1 x_2} & \frac{2}{x_2^2} \end{pmatrix} \geq 0$$

f is convex $\Rightarrow$ quasiconvex



(d) $f(x_1, x_2) = \frac{x_1}{x_2}$ on $\mathbb{R}_{++}^2$

$$\nabla^2 f(x) = \begin{pmatrix} 0 & -\frac{1}{x_2^2} \\ -\frac{1}{x_2^2} & \frac{2x_1}{x_2^3} \end{pmatrix}$$

not positive or negative semidefinite.

$\therefore f$ is not convex or concave.

① f 's sublevel: $L_\alpha = \{(x_1, x_2) \mid f(x_1, x_2) \leq \alpha\} \Rightarrow x_1 \leq \alpha x_2$.

is a halfspace, $\Rightarrow$ ~~convex~~ convex set $\Rightarrow f$ is quasiconvex

② f 's superlevel: $U_\alpha = \{(x_1, x_2) \mid \frac{x_1}{x_2} \geq \alpha\} \Rightarrow x_1 \geq \alpha x_2$

is a halfspace $\Rightarrow$ convex set $\Rightarrow f$ is ~~quasiconvex~~ quasiconcave

(e) $f(x_1, x_2) = \frac{x_1^2}{x_2}$ on $\mathbb{R} \times \mathbb{R}_{++}$

$$\nabla^2 f(x) = \begin{pmatrix} \frac{2}{x_2} & -\frac{2x_1}{x_2^2} \\ -\frac{2x_1}{x_2^2} & \frac{2x_1^2}{x_2^3} \end{pmatrix} = \frac{2}{x_2} \begin{pmatrix} 1 & -\frac{x_1}{x_2} \\ -\frac{x_1}{x_2} & \frac{x_1^2}{x_2^2} \end{pmatrix} = \frac{2}{x_2} \begin{pmatrix} 1 & -x_1/x_2 \\ -x_1/x_2 & (x_1/x_2)^2 \end{pmatrix}$$

main diagonal element ≥ 0 .

$\therefore f$ is convex $\Rightarrow$ quasiconvex

(f) $f(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$, where $0 \leq \alpha \leq 1$, on $\mathbb{R}_{++}^2$:

$$\begin{aligned} \nabla^2 f(x_1, x_2) &= \begin{pmatrix} \alpha(\alpha-1)x_1^{\alpha-2}x_2^{1-\alpha} & \alpha(1-\alpha)x_1^{\alpha-1}x_2^{-\alpha} \\ \alpha(1-\alpha)x_1^{\alpha-1}x_2^{-\alpha} & (1-\alpha)(-\alpha)x_1^\alpha x_2^{-\alpha-1} \end{pmatrix} \\ &= -\alpha(1-\alpha)x_1^{\alpha-1}x_2^{-\alpha} \begin{pmatrix} x_1^{-1} & 1 \\ -x_2^{-1} & -x_2^{-1} \end{pmatrix} \begin{pmatrix} x_1^{-1} & -x_2^{-1} \end{pmatrix} \\ &\leq 0 \end{aligned}$$

f is concave and quasiconcave.



Q3 $f(x, t) = \frac{\|x\|_p^p}{t^{p-1}}$ is convex on $\{x, t \mid t > 0\}$

$$\frac{\partial f}{\partial x_i} = \frac{p |x_i|^{p-1} \cdot \text{sign}(x_i)}{t^{p-1}}, \quad \frac{\partial f}{\partial t} = -\frac{(p-1) \|x\|_p^p}{t^p}$$

$$\frac{\partial^2 f}{\partial x_i^2} = \frac{p(p-1) |x_i|^{p-2}}{t^{p-1}}, \quad \frac{\partial^2 f}{\partial x_i \partial x_j} = 0$$

$$\frac{\partial^2 f}{\partial x_i \partial t} = -\frac{p(p-1) |x_i|^{p-1} \cdot \text{sign}(x_i)}{t^p}, \quad \frac{\partial^2 f}{\partial^2 t} = \frac{p(p-1) \|x\|_p^p}{t^{p+1}}$$

$$H = \begin{bmatrix} \frac{p(p-1) |x_1|^{p-2}}{t^{p-1}} & 0 & \dots & \frac{\partial^2 f}{\partial x_1 \partial t} \\ 0 & \frac{p(p-1) |x_2|^{p-2}}{t^{p-1}} & \dots & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial t} & \dots & \dots & \frac{p(p-1) |x_n|^{p-2}}{t^{p-1}} \end{bmatrix}$$

$p > 1$, main diagonal > 0

H is positive semidefinite

$f(x, t)$ is convex.



Q4 Conjugates.

(a) Max function, $f(x) = \max_i x_i$ on $\mathbb{R}^n$.

$$f^*(y) = \sup_{x \in \mathbb{R}^n} \left\{ \sum_{i=1}^n y_i x_i - \max_i x_i \right\}$$

assume $\max_i x_i = x_k$, $\sum_{i=1, i \neq k}^n y_i x_i = a$, $a \in \mathbb{R}$

$$f^*(y) = a + \sup \{ y_k x_k - x_k \} = a + \sup \{ (y_k - 1) x_k \}$$

① if $y_k - 1 > 0$, $\sup = +\infty$

② if $y_k - 1 < 0$, $\sup = 0$

③ if $y_k - 1 = 0$, $\sup = 0$

$$\therefore f^*(y) = \begin{cases} 0, & y_i \leq 1 \text{ for } i \in [1, n] \\ +\infty & \text{else} \end{cases}$$

(b) Piecewise-linear function on $\mathbb{R}$.

$$f(x) = \max_i (a_i x + b_i)$$

$$\text{break points: } x_i = \frac{b_i - b_{i+1}}{a_{i+1} - a_i} \quad i \in [1, m-1]$$

$$f^*(y) = \sup_{x \in \mathbb{R}} \left[yx - \max_{i \in [1, m]} (a_i x + b_i) \right], \quad y \in [a_1, a_m]$$

$$= -b_i - \frac{(b_{i+1} - b_i)(y - a_i)}{a_{i+1} - a_i}$$



Q6. $f(x) = \sum_{i=1}^n x_i \log \frac{x_i}{1^T x}$

① $f^*(y) = \sup_{x \in \mathbb{R}_+^n} [\langle y, x \rangle - f(x)] = \sup \left[\sum_{i=1}^n y_i x_i - \sum_{i=1}^n x_i \log \frac{x_i}{1^T x} \right]$

Let $\sup s = 1^T x = \sum_{i=1}^n x_i$.

Let $p_i = \frac{x_i}{s}$, so $\sum_{i=1}^n p_i = 1$, $x_i = p_i s$

② $f^*(y) = \sup \left[\sum_{i=1}^n s p_i y_i - \sum_{i=1}^n s p_i \log p_i \right]$

$= \sup_p \left[s \left(\sum_{i=1}^n p_i y_i - \sum_{i=1}^n p_i \log p_i \right) \right]$

$= \sup_{s > 0} s \log \left(\sum_{i=1}^n e^{y_i} \right)$ (Legendre dual of negative entropy)

The $f^*(y)$ is finite only if $\sum_{i=1}^n e^{y_i} < \infty$

So $f^*(y) = \begin{cases} 0, & \text{if } \sum_{i=1}^n e^{y_i} < \infty \\ +\infty, & \text{else} \end{cases}$

